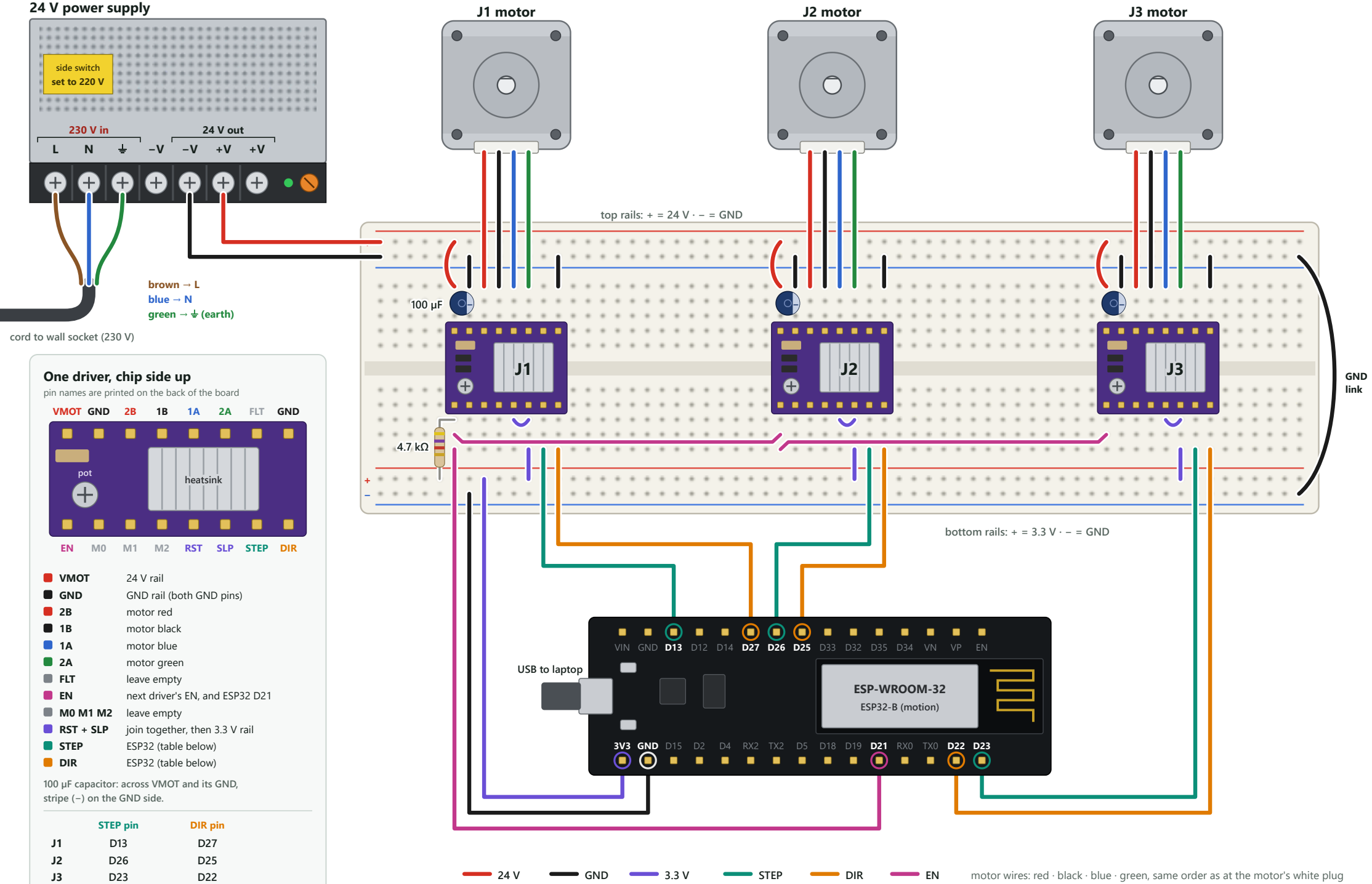
# EasyCliq core wiring

24 V supply · 3 DRV8825 drivers · 3 NEMA17 motors · ESP32-B

**Wire with the plug OUT of the wall.**

Plug motors in or out only with the power off.




## One driver, chip side up

pin names are printed on the back of the board




| | |
|---|---|
| **VMOT** | 24 V rail |
| **GND** | GND rail (both GND pins) |
| **2B** | motor red |
| **1B** | motor black |
| **1A** | motor blue |
| **2A** | motor green |
| **FLT** | leave empty |
| **EN** | next driver's EN, and ESP32 D21 |
| **M0 M1 M2** | leave empty |
| **RST + SLP** | join together, then 3.3 V rail |
| **STEP** | ESP32 (table below) |
| **DIR** | ESP32 (table below) |

100 µF capacitor: across VMOT and its GND, stripe (–) on the GND side.

| | STEP pin | DIR pin |
|---|---|---|
| **J1** | D13 | D27 |
| **J2** | D26 | D25 |
| **J3** | D23 | D22 |

EASYCLIQ · CORE WIRING

# Wire list

Tick each wire as you place it. Plug out of the wall the whole time.

## POWER

| | |
|---|---|
| ☐ Side switch on the supply | set to **220 V** |
| ☐ Cord brown / blue / green | L / N / ⏚ |
| ☐ Supply **+V** | top **+** rail (red) |
| ☐ Supply **−V** | top **−** rail (black) |
| ☐ Top **−** rail | bottom **−** rail (black) |
| ☐ Gap in a rail's red/blue line? | jumper across each gap |

## EACH DRIVER

tick J1, J2, J3

| | |
|---|---|
| ☐☐☐ Driver across the gap, pot bottom-left | VMOT row on top |
| ☐☐☐ **VMOT** column | top **+** rail |
| ☐☐☐ **GND** column next to VMOT | top **−** rail |
| ☐☐☐ **100 µF** across VMOT and that GND | stripe on GND side |
| ☐☐☐ **GND** column next to FLT | top **−** rail |
| ☐☐☐ **RST** column | **SLP** column |
| ☐☐☐ **SLP** column | bottom **+** rail (3.3 V) |
| ☐☐☐ **STEP** | J1 **D13** · J2 **D26** · J3 **D23** |
| ☐☐☐ **DIR** | J1 **D27** · J2 **D25** · J3 **D22** |

## SHARED

| | |
|---|---|
| ☐ J1 **EN** column | J2 **EN** column |
| ☐ J2 **EN** column | J3 **EN** column |
| ☐ J1 **EN** column | ESP32 **D21** |
| ☐ **4.7 kΩ** from J1 EN column | bottom **+** rail (3.3 V) |
| ☐ ESP32 **3V3** | bottom **+** rail |
| ☐ ESP32 **GND** | bottom **−** rail |
| ☐ ESP32 USB | laptop |

## FIRST POWER-ON

1. **Motors not plugged in yet.** Plug the supply in and plug the ESP32 into the laptop. The multimeter should read about 24 V on the top rails and 3.3 V on the bottom rails.
2. **Set each driver to 0.55 V.** Black probe on a GND rail, red probe on the metal top of the driver's pot. 0 V means the driver is asleep: check RST and SLP reach the 3.3 V rail.
3. **Power off and unplug.** Stick a heatsink on each chip. Plug in the motors: red → 2B, black → 1B, blue → 1A, green → 2A.
4. **Power on.** Upload `stepper_test.ino`, open the Serial Monitor at 115200 and type a. All three motors turn once.

**A motor buzzes but doesn't turn:** its two coils are mixed up. Power off, then check that red + black are one pair and blue + green the other.

**Nothing moves at all:** check RST/SLP reach the 3.3 V rail, the GND rails are linked, and the 24 V rail really reads 24 V.